Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Projects

IndicLLM: Transformer Pretraining & Reasoning SFT | *PyTorch, SentencePiece, ONNX Runtime Web*

- Pretrained small GPT-style language models for Hindi and Assamese from scratch in PyTorch, including the data and tokenizers.
- Fine-tuned them for multi-step reasoning, reaching **96%** test accuracy against a 48% baseline.
- Exported the models to run fully in the browser and deployed them as a live demo.

Harpy: Agentic AI Toolkit & Model Context Protocol Server | *Python, MCP, C++, GitHub Actions*

- Built an MCP server that lets AI coding assistants (Cursor, Claude) compile, run and test C++ code locally.
- Added automatic stress testing against a brute-force solution and a sandboxed runner with time limits.
- Published it on PyPI and as a standalone binary, tested in CI across Python versions.

AmbiStory: NLP Plausibility & Disambiguation Engine | *Python, PyTorch, Hugging Face*

- Built a model that rates how plausible each meaning of an ambiguous word is in a short story (SemEval 2026).
- Reached **72%** test accuracy, the best result on the task without using an LLM.
- Combined sentence embeddings with inference features, and used grouped cross-validation so it works on unseen words.

Football Video Analytics with YOLO | *Python, YOLOv8, OpenCV, Streamlit*

- Built a video analysis pipeline that tracks players, referees and the ball with a fine-tuned YOLOv8 model.
- Grouped players into teams by jersey colour and mapped the camera view to the pitch to measure speed and distance.
- Streamed frames in batches to cut memory use from **5.3 GB to 1 GB**, and built a Streamlit app to explore the results.

Technical Skills

Languages: Python, SQL, C/C++, TypeScript, JavaScript.
ML & AI: PyTorch, Hugging Face, Scikit-learn, ONNX, MCP.
Vision & Data: OpenCV, YOLOv8, NumPy, Pandas, Matplotlib.
Tools: Docker, Linux, Git, GitHub Actions, Streamlit.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd by GPA in the B.Tech CSE batch at IIIT Ranchi.
- **PGEE 2025:** Secured All-India Rank 100 in the IIIT Hyderabad Post Graduate Entrance Examination.
- **Megathon 2025:** Placed 3rd in the mobile computing track.

Interests

- Reading web novels and manga.
- Playing chess and badminton.